1> $\left[\left(\frac{2}{3}\right)^{-2} \times \left(\frac{3}{4}\right)^{-3}\right] \div \left(\frac{8}{9}\right)^{-2}$

$= \left[\left(\frac{3}{2}\right)^{2} \times \left(\frac{4}{3}\right)^{3}\right] \times \left(\frac{8}{9}\right)^{2}$

$= \left[\frac{9}{4} \times \frac{64}{27}\right] \times \frac{64}{81}$

$= \frac{16}{3} \times \frac{64}{81}$

$= \frac{1024}{243}$

$= \frac{2^{10}}{3^{5}}$

2>

given: $x + \frac{1}{x} = 6$

$\left(x + \frac{1}{x}\right)^2 = x^2 + \frac{1}{x^2} + 2$

$6^2 = x^2 + \frac{1}{x^2} + 2$

$36 = x^2 + \frac{1}{x^2} + 2$

$x^2 + \frac{1}{x^2} = 36 - 2$

$= 34$

3> given:

workers = 12

Days = 20

working hours per day = 8

New workers = 16

New working hours per day = 6

$$12 \times 20 \times 8 = 16x \times 6$$

$$1920 = 96x$$

$$x = \frac{1920}{96}$$

$$x = 20$$

<u>20 days</u>

4>

given $16x^4 - 81y^4$

$$16x^4 - 81y^4 = (4x^2)^2 - (9y^2)^2$$

$$= (4x^2 - 9y^2)(4x^2 + 9y^2)$$

$$4x^2 - 9y^2 = (2x - 3y)(2x + 3y)$$

Hence, $(2x - 3y)(2x + 3y)(4x^2 + 9y^2)$

6>

Given: principle = 1,25,000

first year = 8%

Second year = 10%

$$A = P\left(1 - \frac{R_1}{100}\right)\left(1 - \frac{R_2}{100}\right)$$

7> given:

$$(3x+4y)^2 - (3x-4y)^2 + (3x-4y)(3x+4y)$$

$$= 4(3x)(4y) + (3x-4y)(3x+4y)$$

$$= 48xy + (3x)^2 - (4y)^2$$

$$= 48xy + 9x^2 - 16y^2$$

9> $r = 7m$, $h = 15m$, $\pi = \frac{22}{7}$

a> $V = \pi r^2 h + \frac{2}{3}\pi r^3$

Cylinder; $= \frac{22}{7} \times 7^2 \times 15 = 2310 m^3$

Hemisphere $= \frac{2}{3} \times \frac{22}{7} \times 7^3 = \frac{2156}{3} m^3$

$\therefore V = 2310 + \frac{2156}{3}$

$= \frac{9086}{3} = 3028.67 m^3$

b> Total Surface Area:

$S = 2\pi rh + 2\pi r^2$

$= 2 \times \frac{22}{7} \times 7 \times 15 + 2 \times \frac{22}{7} \times 7^2$

$= 660 + 308$

$= 968 m^2$

C> Rate = Rs 45 per m^2

Cost $= 986 \times 45$

$= 44370$

10>

Principal = Rs 80,000

Time = 1.5 years = $1\frac{1}{2}$ years

ROI = 10% per annum

a>

$$A_1 = P\left(1+\frac{R}{100}\right)^1\left(1+\frac{R/2}{100}\right)^1$$

$$A_1 = 80000 \times \left(1+\frac{10}{100}\right) \times \left(1+\frac{5}{100}\right)$$

$$A_1 = 80000 \times \left(\frac{11}{10}\right) \times \left(\frac{21}{20}\right)$$

$$= 80000 \times \frac{231}{200}$$

$$= 92,400$$

$$CI_1 = A_1 - P$$

$$= 92,400 - 80000$$

$$= 12,400$$